Angiology
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DOI: 10.1177/0003319717715964
journals.sagepub.com/home/ang
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Increased Pulse Wave Velocity in Systemic Lupus Erythematosus: A Meta-Analysis

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Abstract

Systemic lupus erythematosus (SLE) is associated with increased cardiovascular risk. This study aims to derive a more precise estimation on pulse wave velocity (PWV) level in patients with SLE and related factors. A literature search was performed using PubMed, EMBASE, and The Cochrane Library databases, studies published up to February 28, 2017, in English. Pooled standardized mean difference (SMD) with 95% confidence interval (CI) was calculated by fixed-effect or random-effect model analysis. Heterogeneity test was performed by the *Q* statistic and quantified using I^2; publication bias was evaluated using a funnel plot and Egger's linear regression test. Of 156 studies found, 27 met eligibility criteria, and 14 studies were finally included in the meta-analysis. Meta-analysis revealed that the SLE group had significantly higher PWV levels than the control group; SMD = 0.56 and 95% CI (0.30-0.82). Subgroup analyses showed that body mass index (BMI), sample size, and disease duration were associated with PWV in patients with SLE. Overall, our study suggests that patients with SLE have a higher PWV level, and it is associated with BMI, sample size, and disease duration.

Keywords

systemic lupus erythematosus, pulse wave velocity, meta-analysis

Introduction

Systemic lupus erythematosus (SLE) is one of the commonest connective tissue diseases that span a wide range of clinical features, including accelerated atherosclerosis that have a 4- to 10-fold increased risk of developing cardiovascular (CV) diseases (CVD) compared with the general population.[1-4] Possible explanations for accelerated atherosclerosis in SLE may be due to a high prevalence of conventional risk factors,[5,6] long-term corticosteroid use,[7] the presence of antiphospholipid antibodies,[8] and proatherogenic pathophysiologic phenomena inherent to SLE, all of which lead a chronic low-grade inflammatory state, endothelial dysfunction, and eventually atherosclerotic events.[9]

Arterial stiffness, recognized as a marker of early vascular aging (EVA), can reflect changes in mechanical wall properties that occur early at onset of vascular disease and predispose to major CVD.[10,11] Moreover, increased arterial stiffness has been recognized as an independent risk factor for CVD and measuring it could become an important routine assessment of patients in daily cardiology practice.[12] Arterial stiffness is commonly assessed by metrics of pulse wave velocity (PWV). Pulse wave velocity is defined as the arterial pulse's velocity of moving along the vessel wall and is an indicator of arterial distensibility.[13] Measured PWV is an indicator of arterial wall stiffness and also inversely proportional to arterial dilatation capacity or arterial compliance. Pulse wave velocity is not only a measure of arterial stiffness but also a strong predictor of CV events and all-cause mortality.[14,15] Although PWV levels have been assessed in a number of studies in patients with SLE,[16-19] the results are still not consistent. The present study aimed to determine, through a systematic review and meta-analysis of published reports, a more precise estimation of PWV levels in patients with SLE. In addition, we also explored the related influencing factors of PWV levels in SLE.

Methods

Search Strategy

We performed a prespecified search strategy to identify when PWV first appeared in the literature, including relevant studies in the following databases: PubMed, Embase, and The Cochrane Library database. The keywords were used in all

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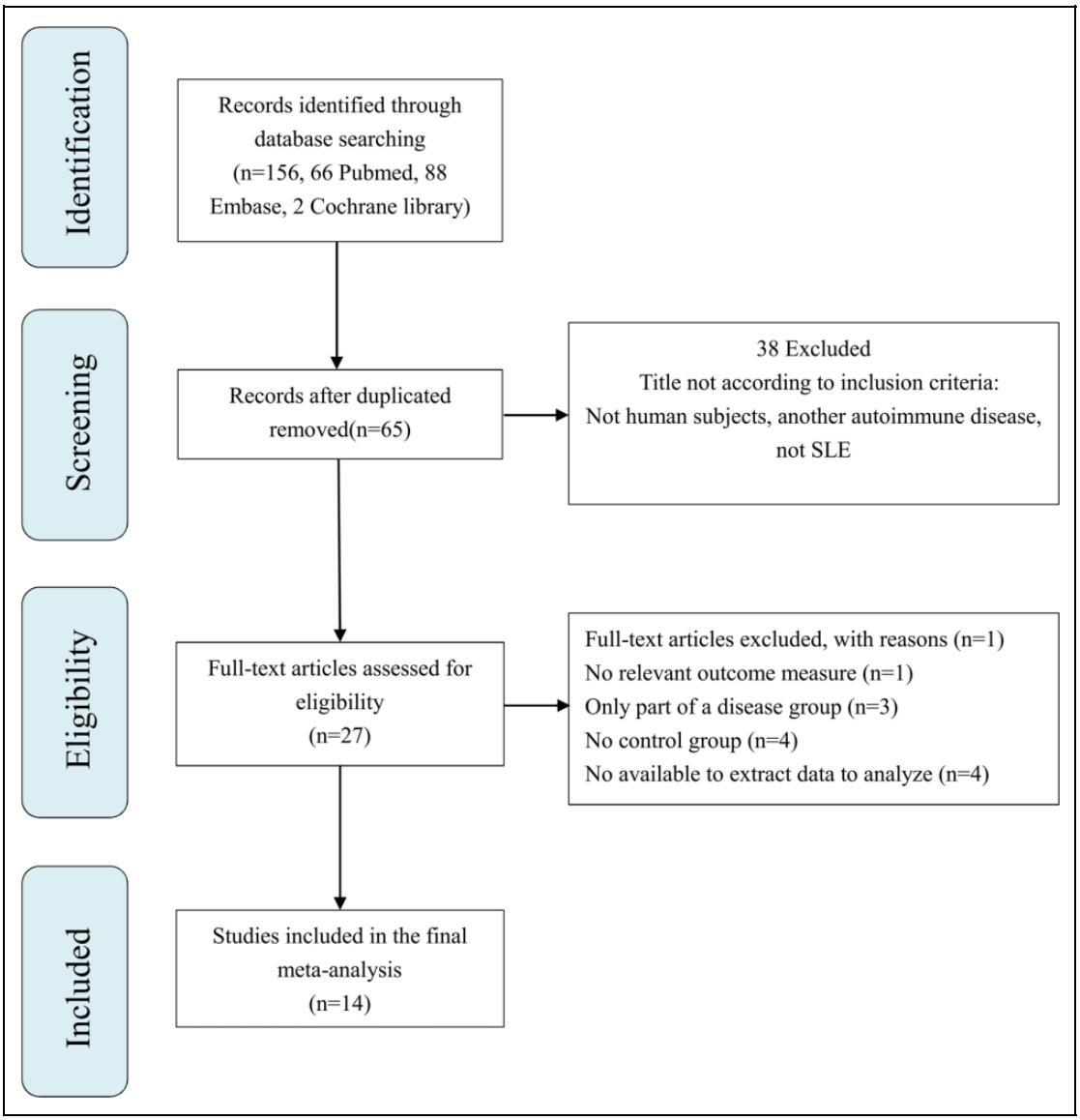

Figure 1. Flow chart of the systematic literature review. Records after duplicated publications removed (n = 65).

fields: "systemic lupus erythematosus" OR "SLE" OR "lupus" AND "carotid-femoral pulse wave velocity" OR "cf-PWV" OR "pulse wave velocity" OR "PWV" OR "aortic PWV" OR "aPWV." All keywords variants were used to increase the number of studies returned by the search. No method or language restrictions were applied, and studies from all countries were eligible. The references in the identified or related articles were manually scanned for other published articles. All resulting articles were then screened for PWV measures and the inclusion of healthy control groups.

Inclusion Criteria and Exclusion Criteria

For inclusion in the meta-analysis, all the following criteria had to be met: (1) case–control, cohort, or cross-sectional studies with data on both patients diagnosed as having SLE and healthy age- and sex-matched controls and (2) provide detailed data about PWV level in both patients with SLE and healthy controls. All patients with SLE had to fulfill the standard accepted diagnostic criteria and healthy controls needed to be clearly described so as to be included in our study. If any study was reported in more than 1 publication, we considered all publications for data abstraction, but only 1 was included in our study. Patients with a history of coronary diseases, infectious diseases, neoplasia, kidney or liver insufficiency, or symptoms suggestive of CVD (angina, arrhythmia, congestive heart failure, stroke, and peripheral arterial disease) were excluded. The exclusion criteria are shown in further detail in Figure 1.

Data Extraction

The quality of the included studies was independently assessed and scored by 2 investigators (Peng Wang and Yan-Mei Mao) using the Newcastle-Ottawa quality assessment scale (NOS)

for observational studies.[20] In the case of disagreement, a third investigator (Li-Na Liu) resolved the disagreement. The mean PWV values (standard deviation) and sample size were extracted, independently with the standard protocol from all articles comparing the PWV of patients with SLE with that of controls. If original important data were unavailable in relevant articles, we e-mailed the corresponding author to obtain the further details.

Statistical Analysis

We extracted the following data from each study, including primary author's name, year of publication, country, sample size of 2 groups, the mean and standard deviation of PWV both in SLE and controls, and other related data. The mean PWV difference (95% confidence interval [CI]) between patients with SLE and control subjects was calculated by using the standardized mean difference (SMD), and its 95% CI was described by a forest plot. The data distribution was assumed to be normal with a z value of $+0.68$ corresponding to the reported 25th and 75th percentiles. In this manner, the standard deviation was calculated.[21] Statistical heterogeneity within studies was examined by the Q statistic.[22] Then, we estimated the effect of heterogeneity in each study by the method, $I^2 = [(Q - df)/Q] \times 100\%$.[23] I^2 values of 25%, 50%, and 75% were qualitatively classified as low, moderate, and high, respectively. If there was no statistical difference about heterogeneity ($P > .05$) among these studies, the fixed-effect model was used. Otherwise, a random-effect model was used for further analyses. Publication bias was visually estimated by a funnel plot, and Egger's regression asymmetry test was performed to explore asymmetry of the funnel plot.[24] If bias was found, the additional tests were applied. To detect the sources of heterogeneity, further subgroup analysis was performed according to body mass index (BMI), sample size, and disease duration. All statistical analyses were conducted by Stata 12.0 software.

Results

Study Characteristics

We identified 156 studies by using electronic and manual search methods. Due to duplicate publication, unclear indicators, unmatched comparisons, review, and conference abstracts, 129 articles were excluded. Of the other 27 potential articles, 14 were finally incorporated in the meta-analysis according to the mentioned criteria. A total of 1752 participants (854 patients with SLE and 898 healthy controls) were ultimately included. The basic characteristics of the selected studies are summarized in Table 1.[16-19,25-34] In 9 studies, patients were diagnosed with SLE by the American College of Rheumatology (ACR) classification criteria, whereas the other 5 studies did not mention the diagnostic criteria. The age, sex, and BMI were matched between patients with SLE and controls in each included study. Newcastle-Ottawa quality assessment scale were determined to assess the quality of included studies (Table 1). Most studies provided high quality as a whole, in which the scores of included studies ranged from 5 to 8 stars.

Heterogeneity Test Results

Statistical heterogeneity among the selected studies was assessed with the χ^2 heterogeneity statistic Q and then quantified with I^2, which measures the inconsistency across study results and describes the proportion of total variation in study estimates that is the result of heterogeneity rather than chance. In this study, the results showed that there was marked heterogeneity among the studies ($I^2 = 84.50\%$, $P < .001$; Figure 2).

Overall Effects and Subgroup Analysis

Overall, through a random-effect model, compared with healthy controls, the analysis showed that patients with SLE had significantly higher PWV (SMD = 0.56, 95% CI: 0.30-0.82; Figure 2). Subgroup analyses showed that BMI, sample size, and disease duration were positively associated with PWV in SLE (Table 2).

Sensitivity Analyses and Publication Bias

A sensitivity analysis compared the meta-analysis results of the studies as a whole to the same meta-analysis with 1 study excluded in each round to determine how robust the findings were. In our study, the results showed no significant change when any 1 study was excluded using random-effect methods (Figure 3). Egger's linear regression test was applied to assess the asymmetry of the funnel plot,[35] and no significant publication bias was found as a whole ($P > .05$; Figure 4).

Discussion

Cardiovascular diseases is a major cause of morbidity and mortality in SLE.[1] Arterial stiffness is a consequence of arterial wall thickening and early changes in mechanical properties of the vascular wall.[36] Previous studies have recognized the link between aortic stiffness and CV events.[12] Aortic PWV is not only a measure of arterial stiffness but also a strong predictor for the future CV events.[37] Furthermore, an abnormal PWV is associated with an increased risk of CVD.[38]

Several studies have evaluated arterial stiffness in patients with SLE. In 41 patients with SLE, arterial stiffness was increased despite a low risk for CVD according to the Framingham score,[18] the study of Bjarnegrad et al[39] in 27 patients with SLE and 27 controls demonstrated that carotid-femoral pulse wave velocity (cf-PWV) was higher in patients with SLE than in controls after correcting for mean arterial pressure and BMI. In addition, the study of Yildiz et al[34] found higher aortic PWV in 24 premenopausal women with SLE compared with 24 age- and sex-matched controls. However, both the study by Tziomalos et al[16] (including 55 patients with SLE and 61 controls) and the study by Quinlan et al[30] (including 45 patients with SLE and 40 controls) found that PWV did not differ between patients with SLE and controls.

Table 1. Characteristics of Abstracted Studies.

		Patients With SLE					Control						
Author, Year	Region	n	Age (Mean [SD]), Years	% Female	Duration, Years	SLEDAI (Means)	n	Age (Mean [SD]), Years	% Female	Aortic PWV Method	Criteria for the Classification of SLE	Study Type	NOS
Tziomalos et al, 2017[16]	Greece	55	53.3 (15.3)	87	NA	NA	61	55.4 (8.1)	92	cf-pwv	NA	Case–control	6
Sabio et al, 2016[25]	Spain	88	35.0 (4.5)	100	NA	NA	102	37.0 (4.5)	100	cf-pwv	NA	Case–control	7
Rivera-Lopez et al, 2016[26]	Spain	93	38.0 (11.3)	100	11.6	2.4	109	37.8 (11.1)	100	cf-pwv	ACR 1997	Case–control	8
Barsalou et al, 2016[27]	Canada	149	17.2 (2.2)	83	3.2	NA	178	14.7 (0.7)	53	cf-pwv	ACR 1982	Cross-sectional	8
Karp et al, 2016[19]	Israel	30	41.2 (13.3)	NA	13.6	6.4	53	40.7 (13.6)	NA	cf-pwv	ACR 1997	Case–control	7
Morreale et al, 2016[28]	Italy	50	38.8 (11.8)	90	NA	NA	20	39.0 (7.9)	90	cf-pwv	ACR 1997	Case–control	7
Sabio et al, 2015[29]	Spain	70	38.0 (4.0)	100	NA	NA	65	42.0 (5.3)	100	cf-pwv	NA	Case–control	7
Quinlan et al, 2015[30]	UK	45	14.0 (0.8)	87	3.3	NA	40	13.2 (4.8)	48	cf-pwv	ACR 1997	Case–control	8
Sabio et al, 2014[31]	Spain	106	36.5 (4.6)	100	NA	NA	101	37.0 (4.6)	100	cf-pwv	NA	Case–control	8
Sacre et al, 2014[18]	France	41	39.0 (10.0)	83	13.0	2.1	35	37.0 (8.0)	80	cf-pwv	ACR 1997	Cross-sectional	5
Sozeri et al, 2013[17]	Turkey	51	14.7 (3.5)	75	9.6	10.1	25	14.9 (3.5)	60	cf-pwv	ACR 1997	Case–control	6
Kocabay et al, 2012[32]	Turkey	22	35.8 (11.1)	95	NA	NA	19	36.2 (15.0)	47	cf-pwv	NA	Case–control	5
Cypiene et al, 2009[33]	Lithuania	30	37.3 (9.2)	NA	8.0	18.4	66	37.5 (8.7)	NA	cf-pwv	ACR 1982	Case–control	7
Yildiz et al, 2008[34]	Turkey	24	33.6 (9.6)	NA	5.3	NA	24	30.2 (9.2)	NA	cf-pwv	ACR 1997	Case–control	6

Abbreviations: ACR, American College of Rheumatology; cf-pwv, carotid to femoral pulse wave velocity; NA, not available; NOS, Newcastle-Ottawa Scale; SD, standard deviation; SLE, systemic lupus erythematosus; SLEDAI, SLE disease activity index; PWV, pulse wave velocity.

In our study, we reviewed 14 comparisons of populations with SLE disease populations (n = 854) versus controls (n = 898). Meta-analysis revealed that there is a statistically significantly greater PWV in patients with SLE compared with controls, supporting the evidence for an increased CVD burden in patients with SLE. Heterogeneity tests showed marked heterogeneity among the studies. Heterogeneity refers to the degree of difference in results among the included studies. Strong heterogeneity suggests that the validity of the meta-analysis is impaired. Egger's tests suggested no publication bias. To minimize the potential influential factor for the results, and to explore the source of heterogeneity, we performed the subgroup analysis. By evaluating the BMI, sample size, and disease duration as subgroups, we found that there were significantly higher PWV levels in different subgroup of patients with SLE, but BMI $\geq$25 group (SMD = 0.37, 95% CI: −0.19 to 0.93), which may be attributed to different treatment regimen and clinical heterogeneity among patients with SLE. For patients with SLE, disease duration $\geq$10 years group, BMI <25, and sample size (n $\geq$ 60) groups, each group individually had higher PWV when compared with controls in the subgroup analysis. This suggested that BMI, sample size, and disease duration may be correlated with the levels of PWV in patients with SLE. Nevertheless, due to different clinical characteristics, sex ratio, and race differences, we could not explore the other possible positive relationships. The potential mechanism of increased CV burden in patients with SLE may be attributed to chronic inflammatory state, where C-reactive protein (CRP) and a number of circulating and endothelium-derived factors, including nitric oxide, von Willebrand factor (vWF), excess of adhesive molecules, and endothelin-1, may influence large artery elasticity providing functional regulation of arterial stiffness.[40-42]

There are several limitations in the current study. First, some of the studies did not have a systemic lupus erythematosus disease activity index (SLEDAI) reported, so we could not directly address aspects of the relationship between PWV and SLEDAI. Second, some studies had small samples from a clinical population or included individuals from only 1 ethnic population and are thus limited in their generalizability. Third, the partial data were approximation numbers; thus, this may be biased compared with the actuality. Fourth, this

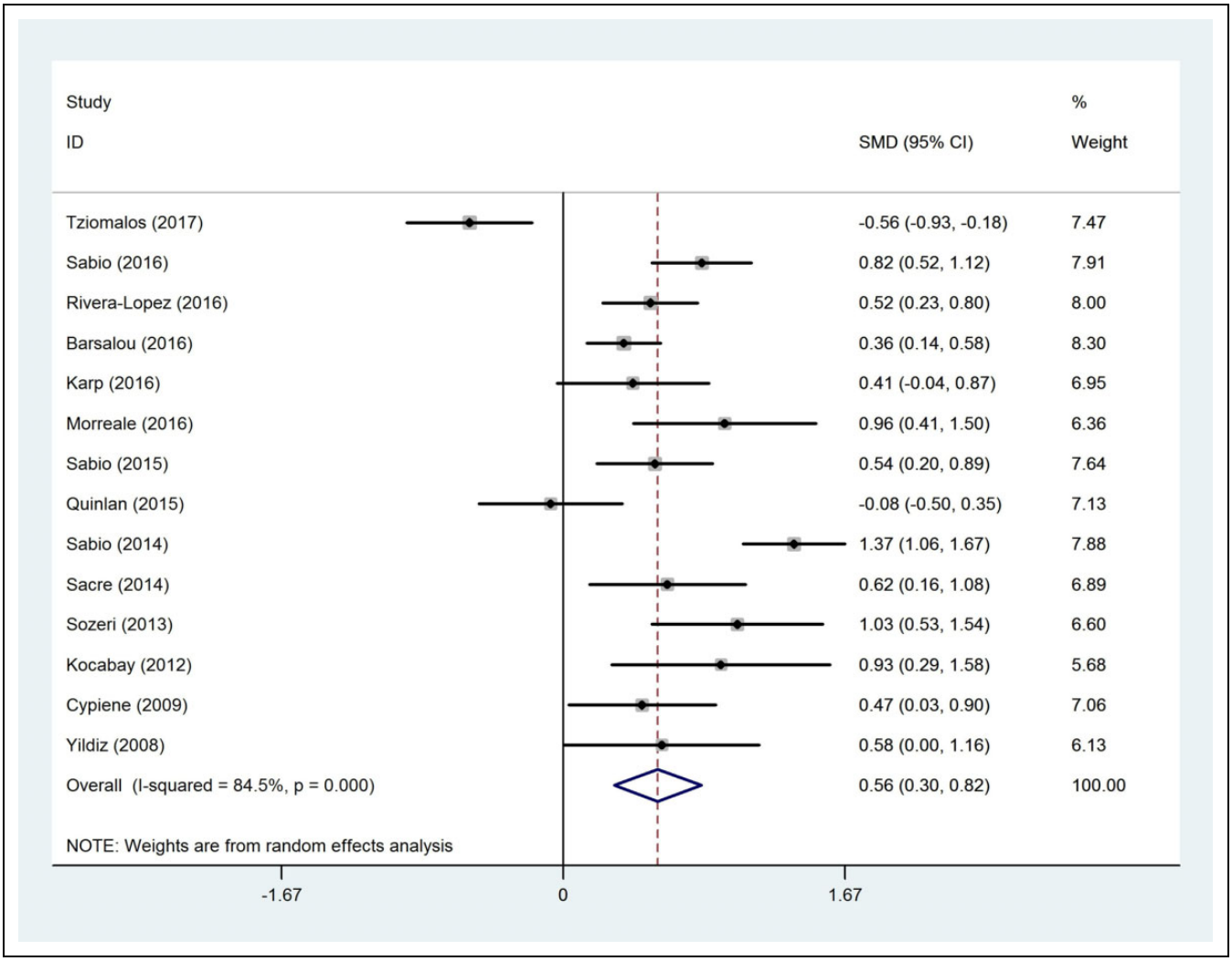

Figure 2. Meta-analysis of 14 studies reporting on PWV in SLE compared with controls. PWV indicates pulse wave velocity; SLE, systemic lupus erythematosus.

Table 2. Subgroup Analysis of PWV Level in SLE.

			Heterogeneity Test			Publication Bias	
Stratification Group	n	SMD (95% CI)	*Q*	*P*	I^2, %	*T*	*P*
Total	14	0.56 (0.30 to 0.82)	84.05	<.001	84.50	0.21	.834
Disease duration							
Duration <10 years	5	0.44 (0.13 to 0.75)	11.41	.022	64.90	1.34	.595
Duration ≥10 years	3	0.52 (0.30 to 0.73)	0.38	.825	0	0.03	.988
Combined	8	0.46 (0.28 to 0.65)	12.55	.084	44.20	0.43	.730
BMI							
BMI <25	7	0.69 (0.32 to 1.05)	37.42	<.001	84.00	−1.27	.788
BMI ≥25	5	0.37 (−0.19 to 0.93)	26.46	<.001	84.90	10.13	.075
Combined	12	0.56 (0.24 to 0.88)	77.88	<.001	85.90	5.40	.061
Sample size							
n < 60	9	0.46 (0.09 to 0.84)	43.81	<.001	81.70	11.34	.009
n ≥ 60	5	0.72 (0.36 to 1.07)	30.83	<.001	87.00	8.67	.349
Combined	14	0.56 (0.30 to 0.82)	84.05	<.001	84.50	0.21	.834

Abbreviations: BMI, body mass index; CI, confidence interval; PWV, pulse wave velocity; SLE, systemic lupus erythematosus; SMD, standardized mean difference.

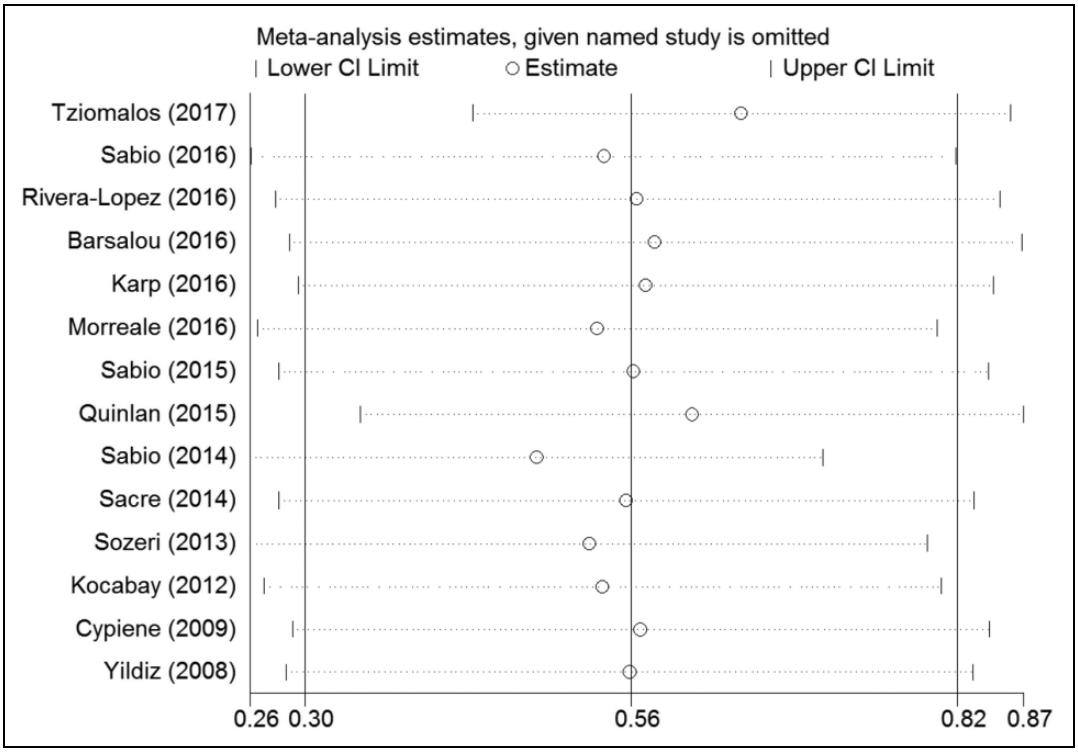

Figure 3. Sensitivity analyses by excluding 1 study at a time.

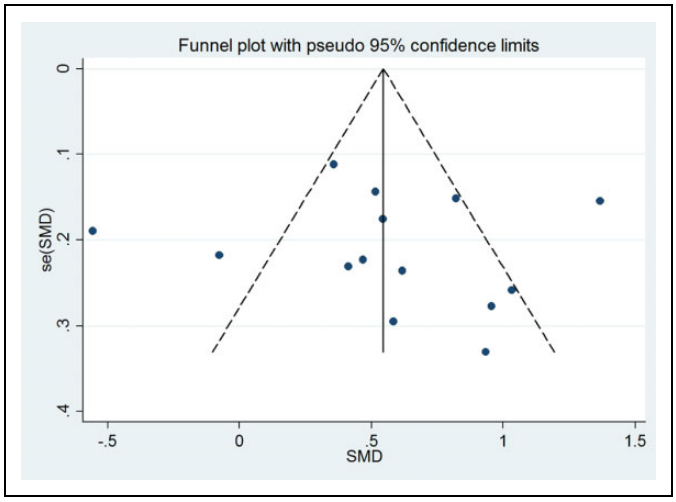

Figure 4. Funnel plot (with pseudo 95% confidence intervals) with the standard error of the PWV difference plotted against the mean difference of PWV of each study. PWV indicates pulse wave velocity.

meta-analysis used a random-effect model which may lead to a more conservative conclusion. Furthermore, this meta-analysis is a retrospective study and could not prove causation which is susceptible to the methodological deficiencies of the pooled studies.

There are, however, particular strengths. Compared with single-country studies, we enhanced generalizability by combining findings from 14 studies across 3 continents. Subgroup analysis was conducted to control the heterogeneity in our study. In addition, no publication bias was detected indicating that the included results may be unbiased. Finally, sensitivity analyses showed that our results were robust.

In summary, we conclude that PWV are higher in patients with SLE compared with healthy controls. In addition, we found that the PWV was influenced by BMI, sample size, and disease duration. However, further analyses with larger samples are still needed to determine the precise relationships between SLE and PWV.

Declaration of Conflicting Interests

The author(s) declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Funding

The author(s) disclosed receipt of the following financial support for the research, authorship, and/or publication of this article: This work was supported by grants from the National Natural Science Foundation of China (81573222).

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